Multi-Structure MORPHEUS Manual

Shared class-correction refinement across related molecular structures

Merlino Development Notes

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Abstract

`MORPHEUS Multi-Structure` is the ensemble-refinement layer for related molecules, conformers or homologous series. Instead of refining every molecule independently, it fits transferable corrections to classes of structural coordinates relative to their own quantum-chemical reference geometries. The workflow is designed for systems where isotopic substitution is sparse for each molecule but chemical class information is shared across the set. This manual describes the model, the TOML job format, class selectors, soft priors, hard constraints, synthon Z_{eff} atom typing, diagnostics and recommended validation workflows.

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1 Purpose

Single-molecule semiexperimental refinement uses isotopologues of one molecule to determine a non-redundant structural model. Multi-structure refinement uses several related molecules to determine shared corrections to chemically equivalent coordinate classes. The absolute structural parameters of different molecules are not forced to be equal. What is shared is the correction to the computational reference for a declared coordinate class.

This is useful for homologous series, conformer pairs and large isotope-poor molecules. For example, two steroid hormones need not have equal C–C bond lengths, but a transferable bias in the computed C–C skeleton class may be estimated from rotational data for both molecules at once. The same idea can separate C–C, C–O single, C=O, angular and torsional classes. C–H distances can be retained at the high-level reference geometry, for example BDPCS3, when the available isotopic data mainly constrain the heavy-atom frame.

2 Theoretical Background

For molecule m and generated coordinate k, the model is

$$q_m^{\mathrm{SE}}(k) = q_m^{\mathrm{QC}}(k) + \Delta_{c(m,k)},$$

where $q_m^{\mathrm{QC}}(k)$ is the quantum-chemical reference value, $q_m^{\mathrm{SE}}(k)$ is the semiexperimental value implied by the ensemble fit, and $c(m,k)$ is the class assigned to that coordinate. The fitted unknowns are the class corrections Δ_c, not the full set of Cartesian coordinates of every molecule.

Each molecule keeps its own geometry, topology, isotopologues, vibrational corrections, symmetry and active coordinate model. The ensemble layer linearizes each molecule around its reference structure and builds one global weighted design matrix for the shared class corrections. If r_m is the single-molecule spectroscopic residual and J_m is the derivative of that residual with respect to the molecule's active coordinates, the class projection C_m maps global corrections into molecule-specific coordinate changes:

$$\delta q_m = C_m \Delta.$$

The global weighted least-squares problem is therefore assembled from blocks

$$r_m + J_m C_m \Delta.$$

Soft priors add Gaussian observations of the form

$$\Delta_c = \Delta_c^{(0)} \pm \sigma_c,$$

whereas hard constraints remove or fix selected corrections. This gives three natural variants: no prior, soft prior and hard constraint. Comparing these variants is a core diagnostic because poorly supported broad classes may be rank-consistent but numerically ill-conditioned.

3 Ensemble Job Format

An ensemble job uses schema:

```
schema = "merlino.semiexp.ensemble.v1"
title = "maleic, phthalic and succinic anhydrides shared class-correction fit"
```

The stable top-level sections are:

Section	Meaning
`[fit]`	Numerical finite-difference step and rank threshold.
`[acceptance]`	Rank, conditioning, support and correlation policy.
`[[molecules]]`	List of ordinary single-molecule MORPHEUS/SE jobs.
`[[classes]]`	Transferable coordinate-class definitions, optional priors and optional synthon selectors.

3.1 Fit and acceptance policy

```
[fit]
step = 1.0e-4
rcond = 1.0e-10

[acceptance]
require_full_rank = true
max_condition_number = 1.0e8
min_residual_degrees_of_freedom = 1
min_molecule_support = 2
high_correlation_review_threshold = 0.98
high_correlation_reject_threshold = 0.9999
```

The result status is `accepted`, `review` or `rejected`. `accepted` means the model passes rank, conditioning, molecule-support and correlation checks. `review` means the model is numerically usable but contains strongly correlated corrections. `rejected` means the class model should not be used without changing classes, priors or atom typing.

3.2 Molecule list

```
[[molecules]]
name = "maleic_anhydride"
job = "../maleic_anhydride/maleic_anhydride_ensemble.mse.toml"

[[molecules]]
name = "phthalic_anhydride"
job = "../phthalic_anhydride/phthalic_anhydride_ensemble.mse.toml"
```

Each molecule points to an ordinary single-molecule job. That job owns its geometry, isotopologues, vibrational corrections, coordinate model and single-molecule options. The ensemble layer does not merge geometries; it merges only the class-correction parameters.

4 Class Selectors

A class must refer to one coordinate type. The implementation rejects class definitions that mix stretches with bends, torsions or out-of-plane coordinates. Selectors can use coordinate kind, atom symbols, value windows, substrings, primitive signatures or synthon Z_{eff} values.

4.1 Atom-symbol selectors

Atom-symbol matching is chemically canonicalized by coordinate type:

- stretches use an unordered atom pair;
- bends preserve the central atom and canonicalize the two terminal atoms;
- torsions use the unordered central-atom pair, i.e. the chemistry of the central bond;
- out-of-plane coordinates use the central atom.

```
[[classes]]
name = "CO_carbonyl"
kind = "stretch"
atoms = ["C", "O"]
value_max = 1.28

[[classes]]
name = "CO_single"
kind = "stretch"
atoms = ["C", "O"]
value_min = 1.28
```

Value windows are evaluated on the reference primitive values and allow broad symbol classes to be split into chemically distinct subclasses.

4.2 Soft priors and hard constraints

Soft priors are declared directly on classes:

```
[[classes]]
name = "CCO_bend"
kind = "bend"
atoms = ["C", "C", "O"]
prior_value = 0.0
prior_sigma = 1.0e-3
```

They regularize weakly supported shared corrections without hiding the spectroscopic residuals. A hard-constraint comparison fixes selected corrections, usually to zero, and removes them from the global variable set. The recommended analysis for a new chemical family is to compare no-prior, soft-prior and hard-constraint variants.

5 Synthon Z_{eff} Typing

The preferred high-resolution atom typing is continuous synthon typing. Each atom is described by an effective atomic number Z_{eff}, and two atoms are considered equivalent when their Z_{eff} values differ by less than a declared threshold. For a primitive coordinate, the chemically relevant ordered signature is used: unordered pair for stretches, central atom retained for bends, central bond for torsions, and central atom for out-of-plane coordinates.

```
[[classes]]
name = "Ccarb_Ocarbonyl"
kind = "stretch"
atoms = ["C", "O"]
synthon_zeff = [5.91, 7.865]
synthon_threshold = 0.035
```

Discrete `synthon_signatures` remain available for reproducibility checks, but thresholded Z_{eff} classes are more transferable across conformers and homologous molecules.

6 Command-Line Workflows

6.1 Production ensemble fit

```
python -m merlino semiexp-ensemble \
  --job examples/semiexp/anhydrides_ensemble/anhydrides_ensemble.mse-ensemble.toml \
  --outdir working/semiexp/anhydrides_ensemble
```

6.2 Prior comparison

```
python -m merlino semiexp-ensemble-compare \
  --job examples/semiexp/anhydrides_ensemble/anhydrides_ensemble.mse-ensemble.toml \
  --outdir working/semiexp/anhydrides_prior_comparison \
  --soft-prior-sigma 1.0e-3
```

This writes no-prior, soft-prior and hard-constraint variants and a summary comparison. It is the preferred diagnostic when deciding whether a class model is chemically informative or merely numerically regularized.

6.3 Synthon threshold scan

```
python -m merlino semiexp-ensemble-synthon-scan \
  --job examples/semiexp/glycine_ensemble/glycine_conformers_synthon.mse-ensemble.toml
    \
  --outdir working/semiexp/glycine_synthon_scan \
  --thresholds 0.015,0.025,0.035,0.050
```

6.4 Paper artifact regeneration

```
python -m merlino semiexp-ensemble-paper \
  --job examples/semiexp/anhydrides_ensemble/anhydrides_ensemble.mse-ensemble.toml \
  --paper-dir doc/papers/ensemble_jpcl \
  --outdir working/semiexp/anhydrides_full_analysis
```

This refreshes CSV/JSON diagnostics, prior-strength scan, leave-one-molecule-out diagnostics and LaTeX fragments for the JPCL-style multi-structure paper.

7 Output Files

Output	Meaning
`ensemble_class_corrections.txt`	Human-readable correction report.
`ensemble_class_corrections.csv`	Class correction, uncertainty and support summary.
`ensemble_class_report.csv`	Per-class diagnostics, priors and acceptance information.
`ensemble_molecule_blocks.csv`	Per-molecule row counts, matched coordinates and residual reduction.
`ensemble_manifest.json`	Scientific manifest for the fitted class model.
`ensemble_covariance.csv`	Covariance matrix for class corrections.
`ensemble_correlation.csv`	Correlation matrix for class corrections.
`run_manifest.json`	Workflow manifest written by CLI entry points.

The prior-comparison workflow also writes `ensemble_prior_comparison` CSV/JSON files and a prior-strength scan. The paper workflow writes the same diagnostics plus generated LaTeX fragments.

8 Diagnostics and Acceptance

The minimum diagnostics for a production model are:

- rank of the weighted design matrix;
- scaled condition number;
- residual degrees of freedom;
- molecule support for each class;
- class-correction standard errors;
- covariance and correlation matrices;
- weighted RMS before and after correction;
- leave-one-molecule-out transfer score when enough molecules are available.

The leave-one-molecule-out score is the held-out WRMS after prediction divided by the uncorrected held-out WRMS. Values below one indicate transferability to the held-out molecule; values near or above one indicate that the class model is not predictive for that molecule.

9 Recommended Regression Set

Current scientific regression targets are:

- maleic, phthalic and succinic anhydrides: no-prior, soft-prior and hard-constraint comparisons;

- glycine conformers I/II: continuous synthon Z_{eff} atom typing and threshold scan;
- primitive signature canonicalization: torsions matched by central bond and out-of-plane coordinates by central atom;
- GUI job-writer round trip for fine selectors, soft priors and absolute paths;
- CLI manifest generation for scientific and workflow manifests.

```
pytest merlino_semiexp/tests/test_ensemble.py \
  gui/tests/test_ensemble_window_helpers.py
```

10 Relation to MORPHEUS

`MORPHEUS Multi-Structure` is an extension of `MORPHEUS`, not a replacement for the single-molecule workflow. Single-molecule MORPHEUS constructs and refines the best model supported by one molecule's isotopic information. Multi-structure MORPHEUS asks a different question: which corrections to a quantum-chemical reference are transferable across a chemically related set? The two workflows share coordinate generation, topology checks, primitive projections, constraints and manifests, but the fitted variables are different.